

Nripesh Pradhan

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Summary

Systems engineer with 7+ years designing and shipping production ML systems, real-time scoring services, and data infrastructure for financial applications. Built the entire quantitative and data stack from scratch for a 10M+ user financial platform across 7+ global markets: classification models, anomaly detection, signal processing pipelines, and real-time decision engines. Mathematics and statistics training at Budapest Semesters in Mathematics.

Experience

Head of Risk Intelligence & Automations, Chipper Cash

Oct 2025 – Present

Lead a quantitative engineering team building ML systems, real-time pipelines, and automated decision engines across 7+ global markets.

- Led a multi-year **systems strategy** to replace third-party vendor platforms with purpose-built quantitative systems, cutting \$500K+/year in costs and reducing end-to-end decision latency by over 80%
- Designed and built a **high-throughput processing platform** from scratch in Python and PostgreSQL, orchestrating 53,000+ events, 20,000+ investigation workflows, and 900+ automated filings across 8 currencies with deterministic audit trails
- Architected a real-time signal detection pipeline evaluating millions of financial transactions against 20+ configurable statistical rule sets, surfacing 2,000+ actionable signals daily with sub-5-minute latency
- Built a **dual-model NLP system** for entity resolution and probabilistic match scoring across millions of records, achieving 85% precision and 95% recall against custom-curated international datasets
- Engineered an autonomous pipeline that ingests unstructured reports via API, applies LLM-based extraction and identity resolution, and executes automated actions in under 20 seconds (down from 30 min), running 24/7

Tech Lead, Risk & Growth, Chipper Cash

Jan 2022 – Oct 2025

Promoted to Staff. Designed and shipped ML-powered scoring systems and production services handling real-time financial data.

- Deployed a **real-time NLP classification service** that parses unstructured transaction data, extracting merchant identity, category, and metadata with sub-second latency at production scale
- Designed and deployed a real-time ML scoring model (**User Onboarding Score**) combining IP geolocation, behavioral timing, device fingerprinting, and identity graph signals. Blocked 6,000+ bad actors/day, saving \$1.4M+/year
- Reduced account takeover events from 30+/day to under 10/quarter by engineering a multi-signal authentication system combining biometric verification, device reputation scoring, and behavioral anomaly detection
- Built a real-time ML scoring service for every card deposit, combining velocity features, rolling statistical aggregates, and behavioral signals at sub-5s latency. Intercepts \$20K+/day in losses
- Automated a dispute engine applying rule-based qualification, generating structured evidence packages, and submitting via processor API. Recovered \$100K+ in disputed funds

Senior Data Scientist, Chipper Cash

Feb 2021 – Jan 2022

Built core financial data infrastructure, statistical analysis frameworks, and production monitoring systems.

- Led company-wide dbt adoption and designed the foundational financial data models powering all analytics, dashboards, and internal tools. Established SQL standards and code review across a 15+ person engineering team
- Conducted quantitative investigations that shaped policy: surfaced referral abuse patterns during a 50K/day user spike, analyzed loss distributions by typology, and identified anomalous return rates, leading to new automated controls
- Built a self-service experimentation platform enabling targeted launches reaching 900K+ users and production alerting monitoring transaction rates across all payment providers in real time

Senior Consultant, Data Science, Logic (acquired by Accenture)

Oct 2018 – Feb 2021

Client-facing quantitative consulting for Fortune 500 retail.

- Delivered predictive modeling and quantitative analytics for **Best Buy** and **Kendra Scott**. Led end-to-end engagements: demand forecasting, inventory optimization, and pricing strategy, presenting to senior leadership

Technical Skills

Languages: Python, SQL, TypeScript **Data Systems:** dbt, Snowflake, PostgreSQL, Airflow **ML/AI:** CatBoost, LLMs, NLP, scikit-learn

Infrastructure: Docker, GCP, AWS, Flask

Quantitative: Pandas, NumPy, statistical modeling **Graph/Vector:** Neo4j, Pinecone **Monitoring:** Datadog, Retool, OpenAI API

Education

Grinnell College, Grinnell, IA

May 2018

B.A. Mathematics and Statistics | **Budapest Semesters in Mathematics** | HBX CORE, Harvard Business School